

Understand Amazon Route 53 Record Types

Summary:-

In this lab, you will explore the commonly used Amazon Route 53 DNS record types. You will create an A Record to map a domain name to an EC2 instance, configure a CNAME Record to create an alias for an existing domain, and review other record types including AAAA, MX, TXT, PTR, SRV, and SPF to understand their purpose and common use cases.

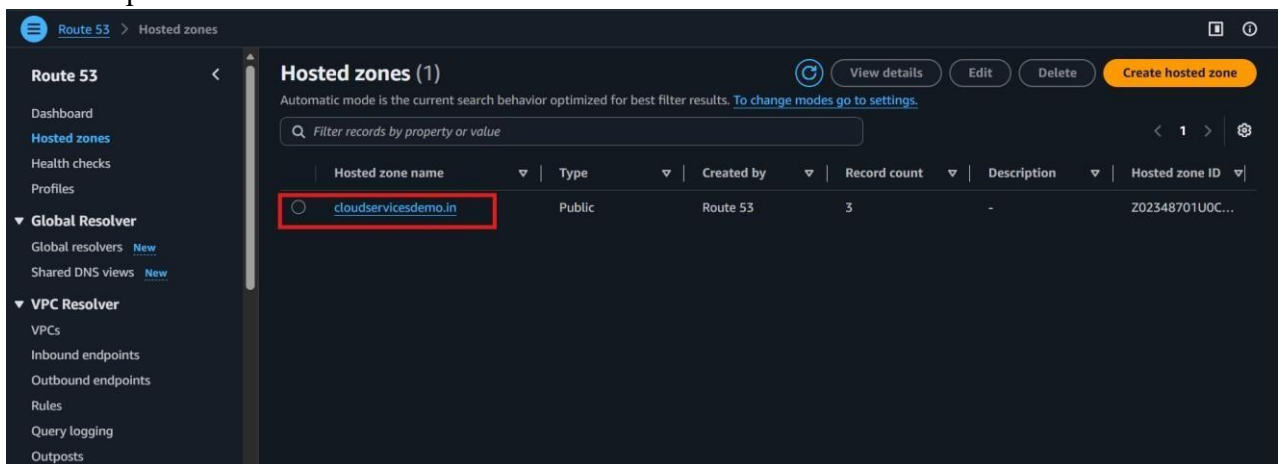
Let's Start:-

- Create an A Record to map a domain name to an EC2 instance.
- Create a CNAME Record for an existing domain.
- Verify DNS record resolution.
- Explore commonly used Route 53 record types.
- Understand the purpose of different DNS records.

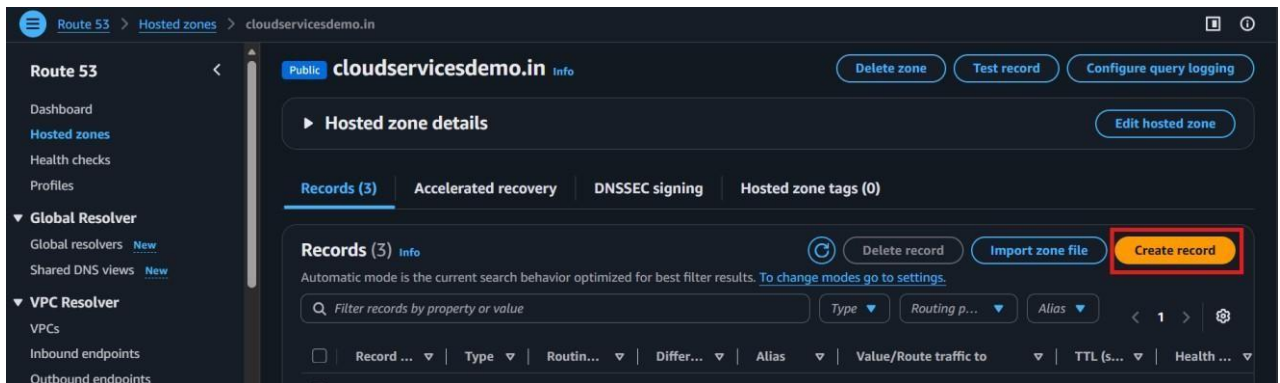
Lab Steps:-

Step 1 – Open the Hosted Zone

- Open the Amazon Route 53 console.



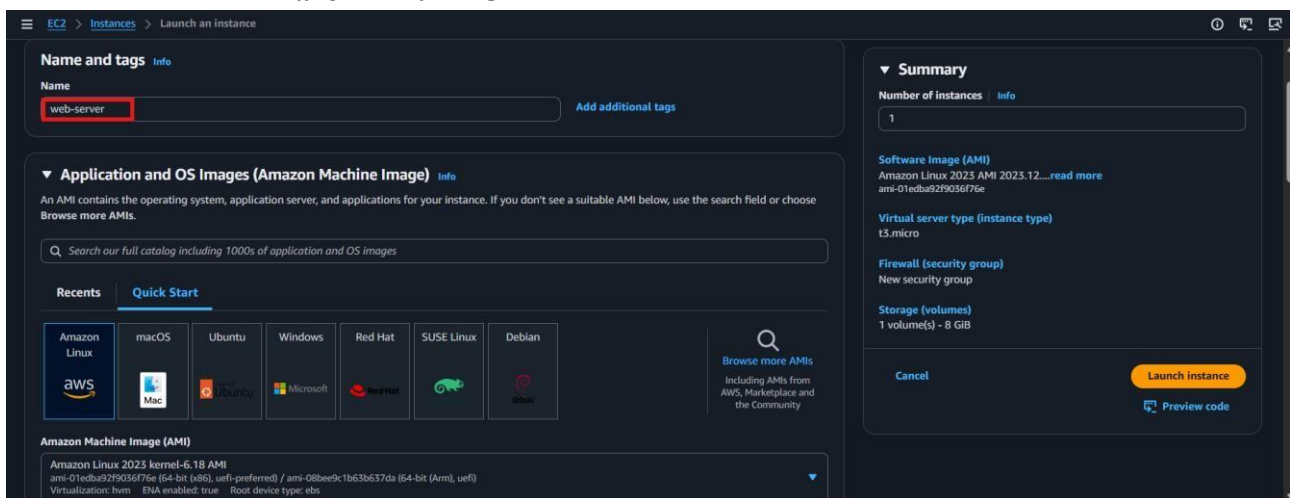
- Select an existing Public Hosted Zone.



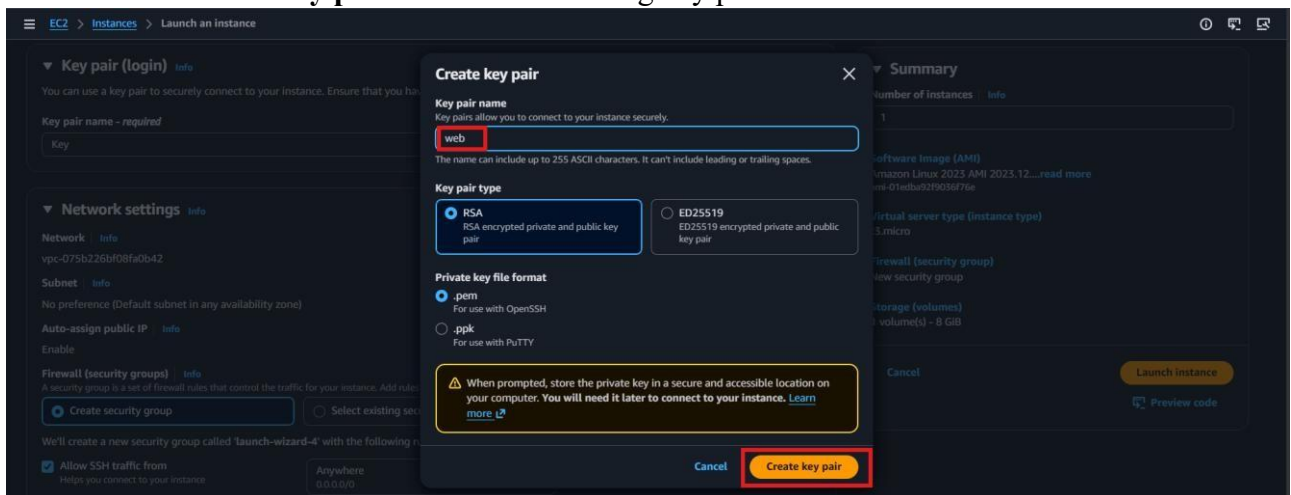
- Click Create Record.

Step 2 – Launch and Configure the EC2 Instance

- Launch an Amazon Linux EC2 instance.



- Create a new key pair or select an existing key pair.



- Allow **HTTP (Port 80)** in the security group.

Key pair name - required: **web** [Create new key pair](#)

Network settings [Info](#) [Edit](#)

Network [Info](#)
vpc-075b226bf08fa0b42

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security group) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-4' with the following rules:

☒ Allow SSH traffic from

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☒ **Allow HTTP traffic from the internet**
To set up an endpoint, for example when creating a web server

Summary
Number of instances: 1 [Info](#)

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12... [read more](#)
ami-01ee8a92f9036f76e

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

- Wait until the instance status is **Running**.
- Connect to the **instance** using **Command Prompt**:
 - `cd <directory-containing-key-pair>`
 - `ssh -i <key-pair>.pem ec2-user@<EC2-Public-IP>`

Instances (1/1) [Info](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[Saved filter sets](#) [Choose filter set](#)

[Instance state = running](#) [Clear filters](#)

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☒	web-server	i-054ca2ac6245f4540	Running	t3.micro	Initializing	us-east-1c	ec2-3-80-71-203.comp...	3.80.71.203

i-054ca2ac6245f4540 (web-server)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID
i-054ca2ac6245f4540

Public IPv4 address
3.80.71.203 | [open address](#)

Private IPv4 addresses
172.31.18.228

IPv6 address
-

Instance state
Running

Public DNS
ec2-3-80-71-203.compute-1.amazonaws.com | [open address](#)


```
[ec2-user@ip-172-31-18-228 ~]$ sudo systemctl start httpd
sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-172-31-18-228 ~]$ sudo tee /var/www/html/index.html > /dev/null <<'EOF'
<!DOCTYPE html>
<html>
<head>
  <title>CloudFront Origin Group Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #232F3E;
      color: white;
      text-align: center;
      padding-top: 100px;
    }
    .container {
      width: 70%;
      margin: auto;
    }
    h1 {
      color: #FF9900;
    }
    .status {
      font-size: 24px;
      color: #00FF7F;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>CloudFront Origin Group Lab</h1>
    <h2>Primary Origin - Amazon EC2</h2>
  </div>
</body>
</html>
```

- Create the web page by running the provided index.html.

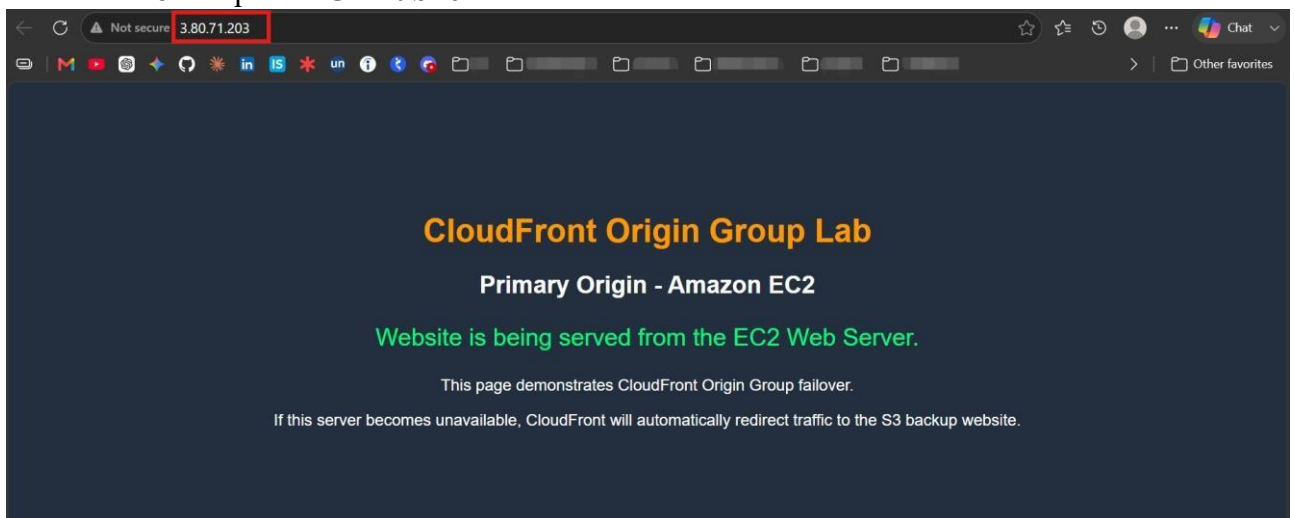
```
sudo tee /var/www/html/index.html > /dev/null <<'EOF'
<!DOCTYPE html>
<html>
<head>
  <title>CloudFront Origin Group Demo</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #232F3E;
      color: white;
      text-align: center;
      padding-top: 100px;
    }
    .container {
      width: 70%;
      margin: auto;
    }
    h1 {
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    }
    .status {
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      color: #00FF7F;
    }
  </style>
</head>
<body>
  <div class="container">
    <h1>CloudFront Origin Group Lab</h1>
    <h2>Primary Origin - Amazon EC2</h2>
  </div>
</body>
</html>
```

```

        font-size: 24px;
        color: #00FF7F;
    }
</style>
</head>
<body>
    <div class="container">
        <h1>CloudFront Origin Group Lab</h1>
        <h2>Primary Origin - Amazon EC2</h2>
        <p class="status">Website is being served from the EC2 Web Server.</p>
        <p>This page demonstrates CloudFront Origin Group failover.</p>
        <p>If this server becomes unavailable, CloudFront will automatically redirect traffic
to the S3 backup website.</p>
    </div>
</body>
</html> EOF

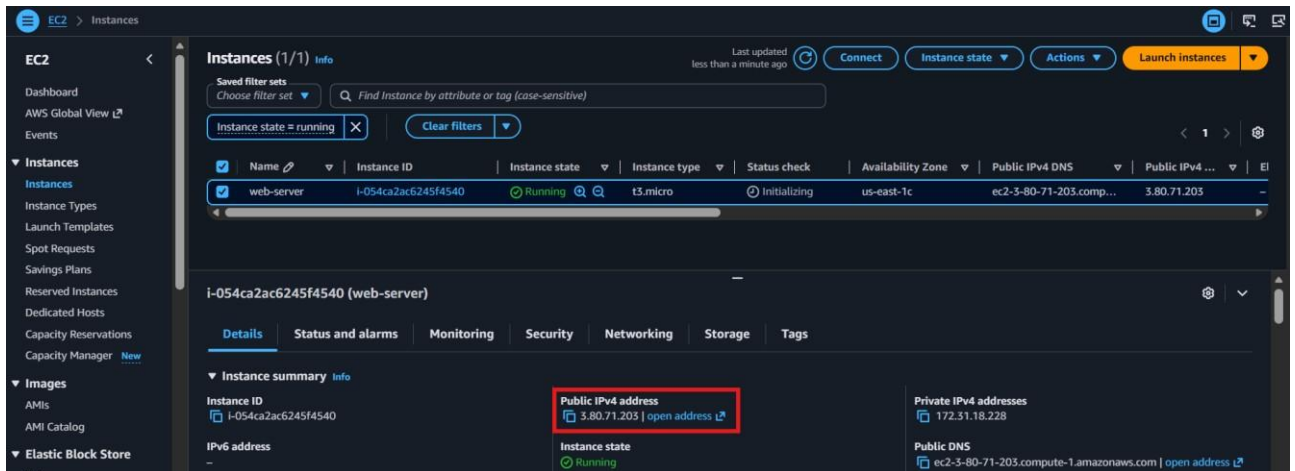
```

-
- Set the appropriate file permissions:
 - `sudo chown apache:apache /var/www/html/index.html`
 - `sudo chmod 644 /var/www/html/index.html`
 - Verify the web server:
 - `curl -s localhost`
 - This step launches and configures the EC2 instance as a web server for the Route 53 DNS record.
 - **Run:**
 - `http://< EC2-Public-IP >`

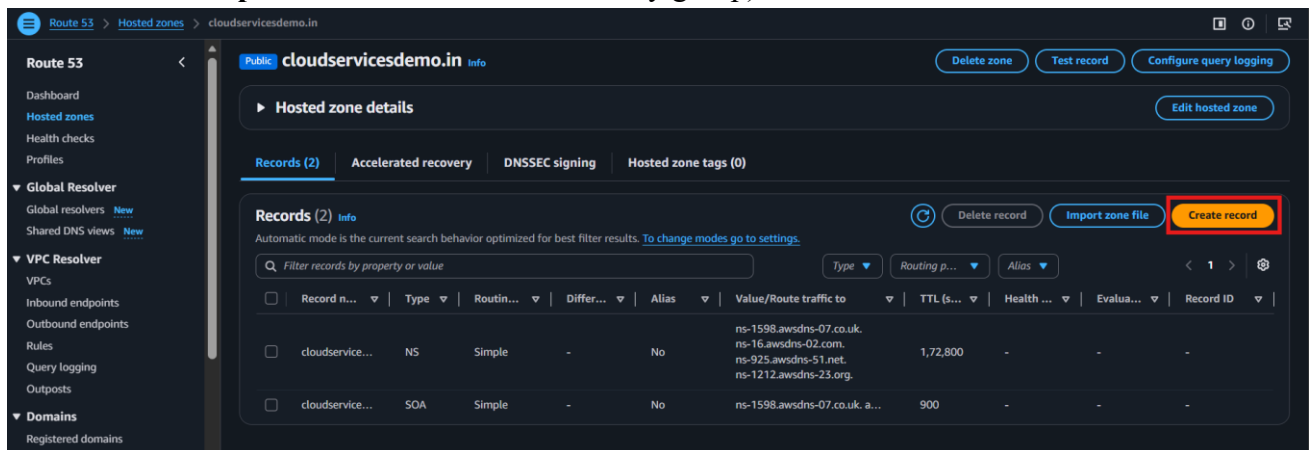


Step 3 – Create an A Record

- Enter the following:
 - **Record Name:** learn
 - **Record Type:** A



- **Value:** Enter the **public IPv4 address of the EC2 instance** (ensure **HTTP traffic on port 80** is allowed in the security group).



- Click **Create Records**.

Route 53 > Hosted zones > cloudservicesdemo.in > Create record

Quick create record Switch to wizard

▼ Record 1 Delete

Record name Info .cloudservicesdemo.in

Record type Info A - Routes traffic to an IPv4 address and some AWS resources

☐ Alias

Value Info

Enter multiple values on separate lines.

TTL (seconds) Info 1m 1h 1d

Recommended values: 60 to 172800 (two days)

Routing policy Info Simple routing

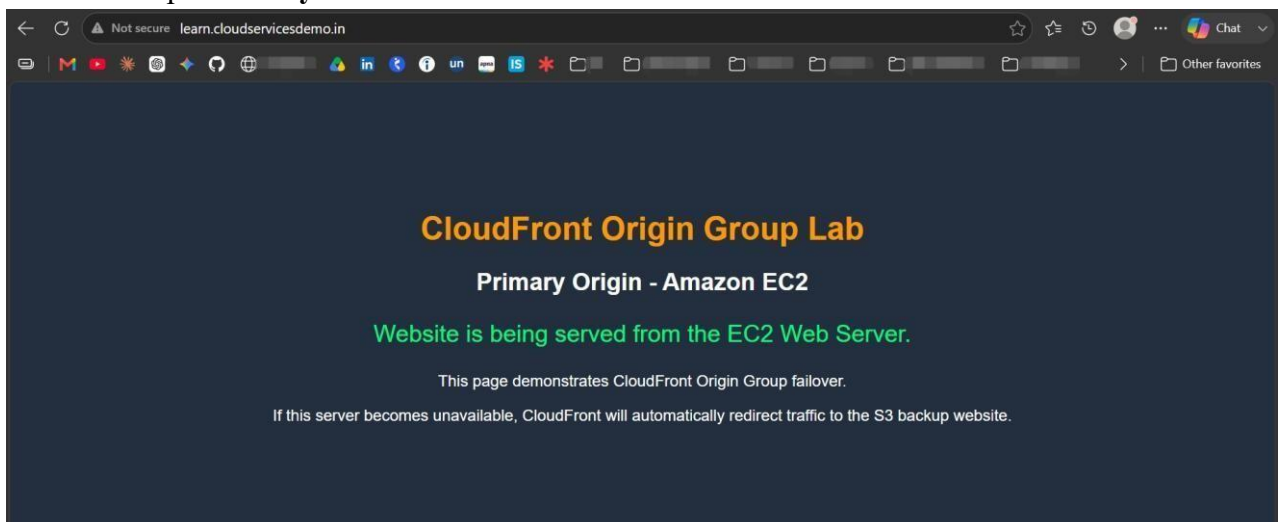
Add another record

Cancel Create records

- Wait a few minutes for DNS propagation if required.

Step 4 – Verify the A Record

- Open a web browser.
- **Enter:**
 - `http://learn.<your-domain-name>`



- Verify that the website loads successfully.
- Alternatively, use **Test Record** in Route 53 to verify that the record resolves to the correct IP address.

Step 5 – Create a CNAME Record

- Click **Create Record**.
- Enter the following:

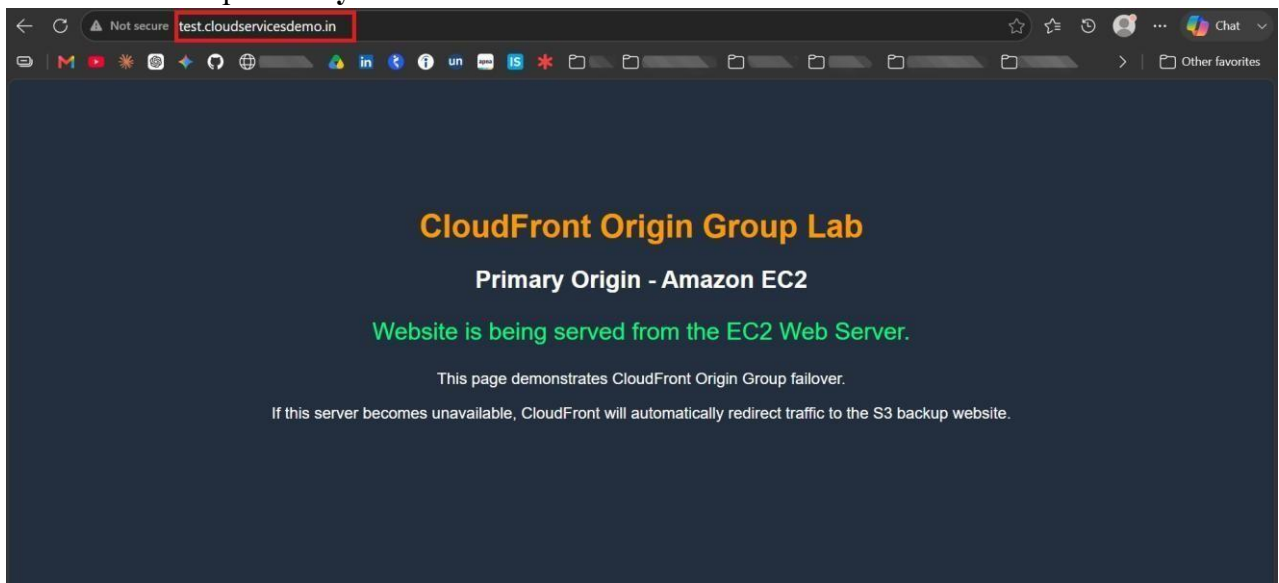
- **Record Name:** test
- **Record Type:** CNAME
- **Value:** learn.<your-domain-name>

The screenshot shows the 'Create record' interface in AWS Route 53. The breadcrumb navigation at the top indicates the path: Route 53 > Hosted zones > cloudservicesdemo.in > Create record. The main heading is 'Create record' with an 'Info' link. Below this is a 'Quick create record' section with a 'Switch to wizard' link. Under 'Record 1', there is a 'Record name' field containing 'test' and a 'Record type' dropdown set to 'CNAME - Routes traffic to another domain name and to some AWS resources'. Below these is an 'Alias' toggle (off) and a 'Value' text area containing 'learn.cloudservicesdemo.in'. At the bottom, the 'TTL (seconds)' is set to '300' and the 'Routing policy' is set to 'Simple routing'.

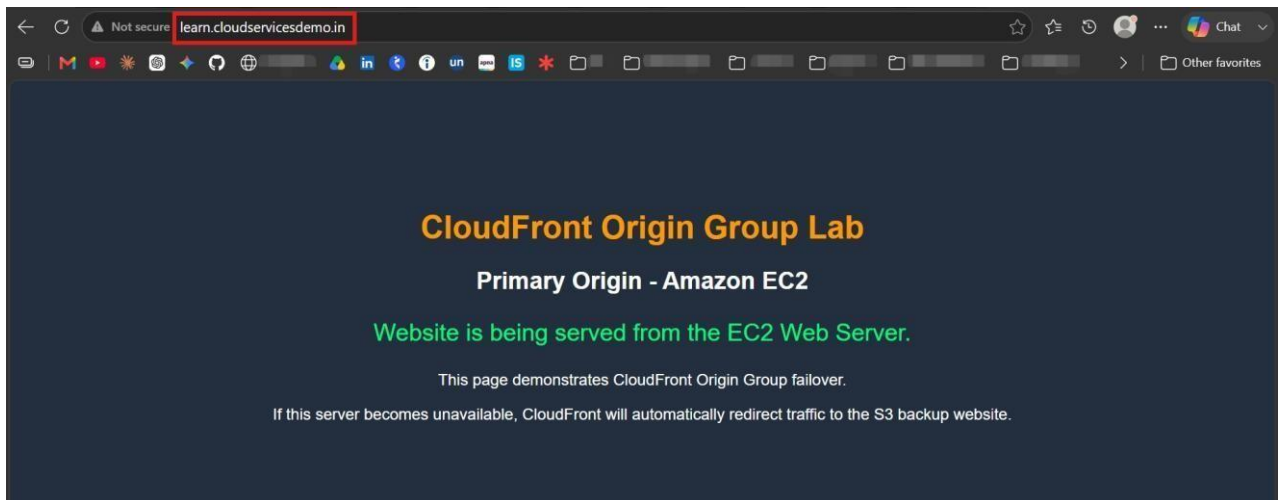
- Click Create Records.

Step 6 – Verify the CNAME Record

- Open a web browser.
- Enter: http://test.<your-domain-name>



- Verify that the request resolves to:
 - http://learn.<your-domain-name>



- Confirm that both domain names open the same website.

Step 7 – Review Other Route 53 Record Types

Review the purpose of the following record types:

- **AAAA Record:** Maps a domain name to an IPv6 address.
- **MX Record:** Specifies the mail server used to receive emails.
- **TXT Record:** Stores text information used for verification and DNS authentication.
- **PTR Record:** Resolves an IP address back to a domain name.
- **SRV Record:** Supports service discovery for applications such as Microsoft Active Directory.
- **SPF Record:** Defines authorized mail servers to help prevent email spoofing.

Verification

- Created an A Record successfully.
- Verified domain name resolution to an EC2 instance.
- Created a CNAME Record successfully.
- Verified that both domain names resolve to the same website.
- Reviewed the purpose of commonly used Route 53 record types.